MINI PROJECT – DAY 10 OF 3

Hospital Data Pipeline

Phase 3: Analysis, Reporting, Archiving

STEP 1 - Python: Full analysis script

```python
# STEP 1 - Python: Full analysis script
import pandas as pd


def load_data(path="processed/enriched_data.csv"):
    return pd.read_csv(path)

def revenue_by_specialty(df):
    total_fee = df.groupby("specialty")["fee"].sum()
    return total_fee.to_dict()

def busiest_doctor(df):
    count = df["doctor_name"].value_counts()
    doctor_name = count.idxmax()
    appointment_count = count.max()
    return doctor_name, appointment_count

def patient_city_summary(df):
    result = df.groupby("city").agg(
        total_appointments = ("appt_id", "count"),
        avg_fee = ("fee", "mean")
    )
    return result

def fee_trend_by_day(df):
    order = ["Monday", "Tuesday", "Wednesday", "Thursday",
             "Friday", "Saturday", "Sunday"]
```

```python
    result1 = df.groupby("day_of_week")["fee"].sum().reindex(order)
    return result1.reset_index(name="total_fee")

def top_patients(df, *args, **kwargs):
    n = kwargs.get("n", 3)

    result2 = (
        df.groupby("patient_name")["fee"].sum().sort_values(ascending=False)
        .head(n)
    )
    return result2

def generate_final_report(df):
    report_lines = []

    rev_spec = revenue_by_specialty(df)
    report_lines.append("\nRevenue by Specialty")
    report_lines.append("----------------------------------------")
    for keys, values in rev_spec.items():
        report_lines.append(f"{keys}: {values}")

    busiest, count = busiest_doctor(df)
    report_lines.append("\nBusiest Doctor")
    report_lines.append("----------------------------------------")
    report_lines.append(f"{busiest} ({count} appointments)")
```

```python
    city_summary = patient_city_summary(df)
    report_lines.append("\nPatient City Summary")
    report_lines.append("--------------------------------------")
    report_lines.append(city_summary.to_string())

    fee_trend = fee_trend_by_day(df)
    report_lines.append("\nFee Trend by Day")
    report_lines.append("--------------------------------------")
    report_lines.append(fee_trend.to_string(index=False))

    top = top_patients(df, n=3)
    report_lines.append("\nTop Patients")
    report_lines.append("--------------------------------------")
    report_lines.append(top.to_string())

    final_report = "\n".join(report_lines)

    with open("reports/fianl_report.txt", "w") as file:
        file.write(final_report)

    report_lines.append("======================================")
    report_lines.append("Pipeline complete. All files archived.")

    return final_report

if __name__ == "__main__":
    df = load_data()
    generate_final_report(df)
```

STEP 2 - Final report format

The final_report.txt must look like this:

python_exercise's > day6_project > hospital_pipeline > reports > fianl_report.txt

```

Revenue by Specialty
--------------------------------------
Cardiology: 1400
Dermatology: 300
General Medicine: 550
Neurology: 1450
Orthopedics: 1150

Busiest Doctor
--------------------------------------
Dr. Carlos Torres (3 appointments)

Patient City Summary
--------------------------------------
         total_appointments  avg_fee
city
Beijing                   1    300.0
Chicago                   1    750.0
Delhi                     2    350.0
Dubai                     1    600.0
Lagos                     1    400.0
Madrid                    1    900.0
Mumbai                    2    525.0
Toronto                   1    150.0

```

STEP 3 - SQL final verification

Run these three queries on freesql.com

and verify results match your Python report:

1. Revenue by specialty (from appointments

   joined with doctors using implicit join).

[SQL Worksheet]*

```sql
-- and verify results match your Python report:
--
-- 1. Revenue by specialty (from appointments
--    joined with doctors using implicit join).
SELECT
    DOC.SPECIALIZATION,
    SUM(AP.FEE) AS TOTAL_REVENUE
FROM APPOINTMENTS AP, DOCTORS DOC
WHERE DOC.DOCTOR_ID = AP.DOCTOR_ID
GROUP BY DOC.SPECIALIZATION
ORDER BY DOC.SPECIALIZATION
-- 2. Patient with highest total fee paid
--     (group by patient_id, sum fee).
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.013 seconds

	SPECIALIZATION	TOTAL_REVENUE
1	Cardiologist	500
2	Dermatologist	550
3	ENT Specialist	650

2. Patient with highest total fee paid

   (group by patient_id, sum fee).

```sql

-- 2. Patient with highest total fee paid
--     (group by patient_id, sum fee).
SELECT
    P.PATIENT_ID,
    SUM(AP.FEE) AS TOTAL_FEE
FROM APPOINTMENTS AP, PATIENTS P
WHERE AP.PATIENT_ID = P.PATIENT_ID
GROUP BY P.PATIENT_ID
ORDER BY TOTAL_FEE DESC
FETCH FIRST 1 ROW ONLY;
-- 3. Count of appointments per day of week.
--    Hint: use TO_CHAR(appt_date, 'Day')
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.006 seconds

	PATIENT_ID	TOTAL_FEE
1	8	900

3. Count of appointments per day of week.

Hint: use TO_CHAR(appt_date, 'Day')

in Oracle SQL.

```sql
-- 3. Count of appointments per day of week.
--    Hint: use TO_CHAR(appt_date, 'Day')
--    in Oracle SQL.
SELECT
    TO_CHAR(APPOINTMENT_DATE, 'DAY') AS DAY_OF_WEEK,
    COUNT(*) AS TOTAL_APPOINTMENTS
FROM APPOINTMENTS
    GROUP BY TO_CHAR(APPOINTMENT_DATE, 'DAY')
    ORDER BY DAY OF WEEK;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.004 seconds

	DAY_OF_WEEK	TOTAL_APPOINTMENTS
1	FRIDAY	2
2	MONDAY	1
3	SATURDAY	1
4	SUNDAY	1
5	THURSDAY	2

STEP 4 - Linux final archiving

1. Confirm reports/final_report.txt exists.

cat reports/final_report.txt

```
cat: reports/final_report.txt: No such file or directory
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ cat reports/fianl_report.txt

Revenue by Specialty
----------------------------------------
Cardiology: 1400
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Orthopedics: 1150

Busiest Doctor
----------------------------------------
Dr. Carlos Torres (3 appointments)

Patient City Summary
----------------------------------------
         total_appointments  avg_fee
city
Beijing                   1    300.0
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Delhi                     2    350.0
Dubai                     1    600.0
Lagos                     1    400.0
Madrid                    1    900.0
Mumbai                    2    525.0
Toronto                   1    150.0

Fee Trend by Day
----------------------------------------
day_of_week  total_fee
     Monday        200
    Tuesday        900
```

2. Create the final delivery archive:

tar -czvf hospital_pipeline_delivery.tar.gz

hospital_pipeline/

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ cd ..
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ tar -czvf hospital_pipeline_delivery.tar.gz hospi
tal_pipeline/
hospital_pipeline/
hospital_pipeline/archive/
hospital_pipeline/mini_projects.ipynb
hospital_pipeline/processed/
hospital_pipeline/processed/appointments_clean.csv
hospital_pipeline/processed/doctors_clean.csv
hospital_pipeline/processed/enriched_data.csv
hospital_pipeline/processed/patients_clean.csv
hospital_pipeline/raw/
hospital_pipeline/raw/appointments.csv
hospital_pipeline/raw/doctors.csv
hospital_pipeline/raw/patients.csv
hospital_pipeline/raw/profiler.ipynb
hospital_pipeline/reports/
hospital_pipeline/reports/analysis.py
hospital_pipeline/reports/cardiology_report.txt
hospital_pipeline/reports/fianl_report.txt
hospital_pipeline/reports/high_fees.txt
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$
```

3. Verify all expected files are inside:

tar -tvf hospital_pipeline_delivery.tar.gz

| grep -E "(.csv|.py|.txt|.json)"

```
 Birth: -
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ tar -tvf  hospital_pipeline_delivery.tar.gz | gre
p -E "(.csv|.py|.txt|.json)"
-rwxrwxrwx prashant/prashant 31245 2026-05-06 13:34 hospital_pipeline/mini_projects.ipynb
-rwxrwxrwx prashant/prashant   657 2026-03-27 21:40 hospital_pipeline/processed/appointments_clean.csv
-rwxrwxrwx prashant/prashant   269 2026-03-27 08:30 hospital_pipeline/processed/doctors_clean.csv
-rwxrwxrwx prashant/prashant  1256 2026-03-28 10:48 hospital_pipeline/processed/enriched_data.csv
-rwxrwxrwx prashant/prashant   324 2026-03-27 07:30 hospital_pipeline/processed/patients_clean.csv
-rwxrwxrwx prashant/prashant   465 2026-03-23 22:23 hospital_pipeline/raw/appointments.csv
-rwxrwxrwx prashant/prashant   206 2026-03-23 22:22 hospital_pipeline/raw/doctors.csv
-rwxrwxrwx prashant/prashant   256 2026-03-23 22:20 hospital_pipeline/raw/patients.csv
-rwxrwxrwx prashant/prashant 66667 2026-04-03 12:16 hospital_pipeline/raw/profiler.ipynb
-rwxrwxrwx prashant/prashant     0 2026-04-03 11:23 hospital_pipeline/reports/analysis.py
-rwxrwxrwx prashant/prashant   141 2026-03-28 22:10 hospital_pipeline/reports/cardiology_report.txt
-rwxrwxrwx prashant/prashant  1083 2026-05-06 13:00 hospital_pipeline/reports/fianl_report.txt
-rwxrwxrwx prashant/prashant   104 2026-03-28 21:39 hospital_pipeline/reports/high_fees.txt
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$
```

4. Check total size:

du -sh hospital_pipeline_delivery.tar.gz

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ du -sh hospital_pipeline_delivery.tar.gz
20K     hospital_pipeline_delivery.tar.gz
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ df -sh hospital_pipeline_delivery.tar.gz
df: invalid option -- 's'
Try 'df --help' for more information.
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ df -h hospital_pipeline_delivery.tar.gz
Filesystem      Size  Used Avail Use% Mounted on
D:\             122G  1.4G  120G   2% /mnt/d
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$
```

5. Set final permissions:

All .py files → chmod 755

All .csv files → chmod 644

All .txt files → chmod 644

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ find . -type f -name "*.py" -exec chmod 755 {} \;
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ find . -type f -name "*.csv" -exec chmod 644 {} \
;
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ find . -type f -name "*.txt" -exec chmod 644 {} \
;
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$
```

6. Print a final confirmation:

echo "Pipeline delivery package ready."

echo "Files inside:"

tar -tvf hospital_pipeline_delivery.tar.gz

| wc -l

echo "files packaged."

```
rashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ echo "==== Pipeline Delivery Package Ready. ===="

cho "Files Inside:"
ar -tvf hospital_pipeline_delivery.tar.gz
ar -tvf hospital_pipeline_delivery.tar.gz | wc -l
cho "Files Packaged."
=== Pipeline Delivery Package Ready. ====
iles Inside:
rwxrwxrwx prashant/prashant 0 2026-03-29 09:07 hospital_pipeline/
rwxrwxrwx prashant/prashant 0 2026-03-23 22:12 hospital_pipeline/archive/
rwxrwxrwx prashant/prashant 31245 2026-05-06 13:34 hospital_pipeline/mini_projects.ipynb
rwxrwxrwx prashant/prashant     0 2026-03-28 10:44 hospital_pipeline/processed/
rwxrwxrwx prashant/prashant   657 2026-03-27 21:40 hospital_pipeline/processed/appointments_clean.csv
rwxrwxrwx prashant/prashant   269 2026-03-27 08:30 hospital_pipeline/processed/doctors_clean.csv
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rwxrwxrwx prashant/prashant 66667 2026-04-03 12:16 hospital_pipeline/raw/profiler.ipynb
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rwxrwxrwx prashant/prashant   141 2026-03-28 22:10 hospital_pipeline/reports/cardiology_report.txt
rwxrwxrwx prashant/prashant  1083 2026-05-06 13:00 hospital_pipeline/reports/fianl_report.txt
rwxrwxrwx prashant/prashant   104 2026-03-28 21:39 hospital_pipeline/reports/high_fees.txt
8
iles Packaged.
rashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project$ |
```